Early Warning Systems for Banking Crises

Evaluating the Interpretability–Performance Trade-off Under Structural Non-Stationarity

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10-min presentation

The Problem: A Known Failure, Unknown Mechanism

What the literature established before this dissertation

What we knew

- ML models (RF, XGBoost) outperform logistic regression under expanding-window CV
- Under strict temporal validation, all models fall below AUROC 0.75 benchmark (Beutel et al., 2019)
- Extending training data did not resolve the problem
- Sentiment from central bank speeches provides directional signal

What was missing

- No mechanism-level explanation of why failure occurs
- No test of whether interpretable models are more or less vulnerable
- No clean separation of algorithm effect from sentiment effect
- SHAP is approximate — cannot distinguish model error from attribution error

Data, Design and Model Sequence

JST Macrohistory R6 · 18 economies · 1985–2020

569	37	26	16,622	41+14
country-year observations	crisis events (Stage 1, 6.5%)	crisis events (Stage 2, 8.15%)	BIS speeches FinBERT-processed	macro + sentiment features

Model Sequence

M1	M2	M2.5 EBM	M3
Logistic Regression 19 features (lag1 only)	Random Forest 41 macro features	Explainable Boosting 41 macro features ★	XGBoost + FinBERT 55 features (macro+sent)

★ *Novel contribution — natively interpretable non-linear model*

Main Results: EBM Leads In-Sample, Collapses Prospectively

The central empirical paradox

Stage 1 — Expanding-Window (1988–2020)

Model	AUROC	AUPRC
M1: Logistic	0.606	0.139
M2: Random Forest	0.636	0.224
M2.5: EBM ★	**0.753**	**0.273**

Stage 2 — Expanding-Window (2003–2020)

Model	AUROC	AUPRC
M2-matched: RF	0.939	0.343
M2.5-matched: EBM ★	**0.960**	**0.532**
M3: XGBoost+FinBERT	0.932	0.411

⚠️ Strict Temporal Validation — Split B (Train: 1988–2002, Test: 2003–2011)

M2: Random Forest

AUROC 0.481

Below random (0.5)

M2.5: EBM ★

AUROC 0.333

Severely below random — worse than RF

Why EBM Fails: Shape Function Regime Instability

Figure 4.7 — the dissertation's central analytic contribution

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Panel logistic confirms: term_spread_lag1 $\beta = -0.646$ ($p=0.004$) vs lag3 $\beta = +0.630$ ($p=0.003$) — sign reversal confirmed parametrically from a completely independent framework.

Same feature, opposite risk

At high term_spread_lag3 values: pre-GFC model assigns REDUCED risk; GFC-inclusive model assigns INCREASED risk. The curves cross.

Not an approximation error

EBM shape functions ARE the model. No SHAP approximation. Instability = property of the data-generating process, not the attribution tool.

Direct mechanism

This sign reversal explains AUROC 0.333 mechanically: model applies pre-GFC mappings to GFC data → systematic misclassification.

Sentiment and Ensemble: Directional but Unstable

H1, H2 and H3 verdicts

H1	**Does FinBERT sentiment improve AUROC/AUPRC vs macro baseline?** M3 AUPRC 0.411 > M2-matched 0.343. But CI [−0.130, +0.312] spans zero. EBM macro-only AUPRC 0.532 > M3 — algorithm effect dominates.	**Directionally supported**
H2	**Does at least one FinBERT feature rank top-5 by SHAP?** No sentiment feature in top 20 EBM native importance. SHAP aggregate: macro features 73.8% of total attribution. Convergence = shared sample correlation, not causal signal.	**Not supported**
H3	**Is there a quantifiable accuracy-interpretability trade-off?** Trade-off is not binary. EBM leads in-sample. Under prospective evaluation, EBM fails more severely — flexibility enables interpretability AND regime-specific overfitting. Same mechanism.	**Supported and refined**

Ensemble E1-EBM: AUROC 0.955 · Brier 0.056 (best calibration). AUPRC 0.433 < EBM standalone 0.532 — probability dilution from averaging with lower-AUPRC M2-matched.

Conclusions: What This Dissertation Establishes

Four findings, one structural argument

1. More data does not fix the problem

15 additional years of crisis history, including the Nordic cluster, does not resolve GFC generalisation failure. Regime discontinuity is structurally too large for aggregate macro features to bridge.

2. Interpretability does not confer robustness

EBM leads in-sample but fails most severely prospectively. The flexibility that enables shape function interpretability is the same mechanism that produces regime-specific overfitting.

3. EBM makes the failure mechanism visible

Shape function instability at term_spread_lag3 directly explains the AUROC collapse. Static EWS models may be conceptually misaligned with structural break environments.

4. Sentiment is directional but unstable

FinBERT contributes a GFC-episode-specific signal. Algorithm choice explains at least as much of the Stage 2 performance difference as the sentiment extension.

Overarching argument: ML-based EWS failure is not a function of model choice, feature design, or data availability — it is structural instability in the mapping between predictors and crisis outcomes across regimes.

Implications and Close

What should change in EWS design — and what comes next

Practical implications for EWS design:

- A single static classifier is insufficient. Operational EWS requires a regime-classification layer or portfolio of mechanism-specific models.
- EBM should be adopted for its diagnostic value — shape functions reveal when a model's learned mapping has shifted, providing regulators a direct signal to apply scepticism.
- Ensemble averaging is most defensible for calibration (Brier 0.056) rather than discrimination — useful for probability thresholds in policy review.
- Sentiment augmentation requires validation across structurally distinct crisis episodes before operational deployment. Current evidence is hypothesis-generating, not confirmatory.

Future research

- Regime-conditional models (EBM-R1/R2) — contingent on sufficient crisis episodes per regime
- Cross-country contagion features (network linkages, Cetorelli & Goldberg 2011)
- Topic-disaggregated sentiment — credit, monetary policy, corporate channels
- Bayesian updating framework for real-time regime detection

The central limitation of EWS is the absence of a stable mapping between predictors and crisis outcomes across regimes — ML makes this visible; it cannot resolve it.

Q&A — Anticipated Challenging Questions

5-minute response window · Be concise and direct

Q1 Split B has only 6 training positives. How can you draw conclusions from it?

Q2 EBM 0.333 could be sample thinness, not regime shift. How do you distinguish?

Q3 You cannot separate sentiment signal from algorithm effect in Stage 2. Is H1 meaningful?

Q4 Your expanding-window AUROC figures inflate due to GFC concentration. Are Stage 2 numbers informative at all?

Q5 Why not use a regime-switching model to directly address non-stationarity?

Q6 SHAP already shows feature instability across time — what does EBM add that SHAP cannot provide?